USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Non LTTPR & LTTPR Transparent – Link Training Limitation
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Link training in Non LTTPR or LTTPR Transparent mode with an SSC turned on and an LTTPR between the DPTX and the DP IN Adapter is not supported by the DP Tunnel.

Benefits as a result of the changes:
Spec is more complete

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

Actual Change

(a). Section 10.3.1 DisplayPort

10.3 Interfaces

10.3.1 DisplayPort

A DP Adapter shall support three modes of operation:

- LTTPR Non-Transparent – LT-tunable PHY Repeater (Non-Transparent Mode).
- LTTPR Transparent – LT-tunable PHY Repeater (Transparent Mode).
- Non-LTTPR – Non-LT-tunable PHY Repeater.

DisplayPort link training is executed over the tunnel in one of three ways:

- DPTX Managed, Sequential Link Training – In this method, the DPTX is aware and manages the link training of the two DisplayPort links: (1) the link between the DPTX and the DP IN Adapter and (2) the link between the DP OUT Adapter and the DPRX. Link training is done sequentially, where the upstream link (1) is trained first and the downstream link (2) is trained second. This method of link training is used when operating in LTTPR Non-Transparent mode. See Section 10.4.10.1.

- Autonomous, Concurrent Link Training – In this method, the DPTX does not manage link training between the DP OUT Adapter and the DPRX. Instead, it manages only the link training to the DP IN Adapter. The DP OUT Adapter manages the link training to DPRX in parallel to the link training between the DPTX and DP IN Adapter. The DP IN Adapter indicates upstream link training completion only after it is notified of the downstream link training completion by the DP OUT Adapter, ensuring the end-to-end link training completion by the time the DPTX exits the link training sequence. This method of link training is used when operating in Non-LTTPR and LTTPR Transparent modes. See Section 10.4.10.2.

  Note: DP Tunneling may not operate correctly when conducting DisplayPort Link training in Non-LTTPR and LTTPR Transparent modes, ~~is not supported~~ in case all the below conditions are met:

  - *At least one LTTPR is located between the DPTX and the DP IN Adapter.*
  - *The DPTX applies SSC.*

  The above limitation is due to a potential underrun at the DP OUT Adapter buffer which starts storing data on a non-SSC clock and later transitions to an SSC clock.

- DPTX Managed, Concurrent Link Training – In this method, the DPTX manages the overall link training process, which consists of link training initiation, transitioning between the different link training phases, and link training completion. The EQ DONE and CDS phases are performed concurrently and managed by each DFP per DP link. This method of link training is used when operating in 128b/132b LTTPR mode. See Section 10.4.10.3.